

NCTU-EE IC LAB – SPRING 2016

Final Online Test - Quiz 1:00p.m. ~ 2:00p.m.

1. (1)Mention two advantages and two disadvantages about both asynchronize reset and synchronize reset.

asynchronize:

advantages: Less area,reset is immediate

disadvantages:noisy reset line cause unwanted reset,Metastability

synchronize:

advantages: glitch filtering,easier work with cycle based simulator

disadvantages:large area,can't work without clock signal

- (2) Try to explain the "setup time" & "hold time", and give two methods to fix setup time violation and one method to fix hold time violation.

Setup time is the time that a signal should arrived and become stable at the cell before the clock triggered. Hold time is the time that a signal should retain its value after the clock triggered. Both parameters can ensure that the correct data are latched correctly.

fix setup time violation: pipeline,parallel circuit,larger clock cycle time

fix hold time:adding buffer

- (3)Explain what is different between sequential circuit and combinational circuit. And explain why we use FSM.

sequential circuit:

It is constructed by previous value which stored by memory elements like regs or RAM and immediate value that input right now.And sequential circuit is control by clk signal.

combinational circuit:

It depends only on immediate values that are inserted right now. And output signals response immediately.

FSM:

We use FSM to design our states,by doing that,we use this centor controller to better manage our whole circuits.We divide control signals and data flow.

2. (1) What is the difference between RTL simulation and Gate-level simulation?

In the RTL level we only care about whether it is cycle accurate or not, there is no detail timing information in this level.

(2) What is the meaning of the following command, in what situation will we need to add them?

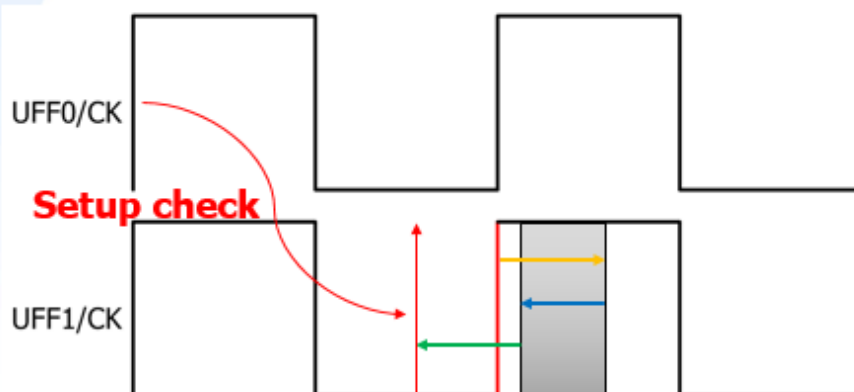
```
//synopsys translate_off  
//synopsys translate_on
```

It tells the Design Compiler do not compile the files between these statement. When we use vendor supply hardmacro or designs from others we may add this statement. The reason is that these modules are already optimized and verified so that we do not have to resynthesize them.

3. (1) Please describe what is STA
(2) Please describe what is DTA
(3) Compared STA and DTA, which one is more efficient? why?
(1) STA analyzes every timing paths in design according to constraints. STA will check timing path between any two FFs.
(2) DTA uses input test vectors to do gate-level simulation
(3) In large design, STA will be more efficient, because in DTA, the growing size of test vectors increase simulation time
4. Please describe how EDA tools calculate the setup check and hold check. Label the waveform below by **ideal_clock_rise_edge**, **rise_delay**, **setup_uncertainty**, **hold_uncertainty**, **Tsetup**, **Thold**.

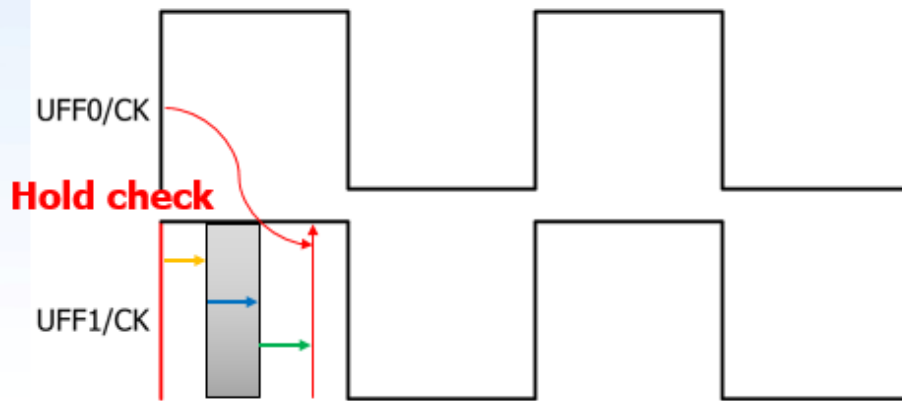
– Setup check

- check_edge : ideal_clock_rise_edge
+ rise_delay(network latency+source latency)
– setup_uncertainty(minus uncertainty)
– T_{setup}(library setup time)



– Hold check

- check_edge : ideal clock rise edge
 + rise_delay(network latency+source latency)
 + hold_uncertainty(plus uncertainty)
 + T_{hold} (library hold time)



- Briefly describe the different circuit between if-then-else and case after synthesis?
 If-then-else: a cascaded encoder
 case: a single-level mux
- Please finish the waveform according to the Figure 1 and the timing setting below.

```

not #(0.5, 0.5) n0( seln, sel ) ;
and #(0.5, 1.0) a0( w0, a, seln );
and #(0.5, 1.0) a1( w1, b, sel );
or  #(1.0, 0.5) o0( out, w0, w1 ) ;
  
```

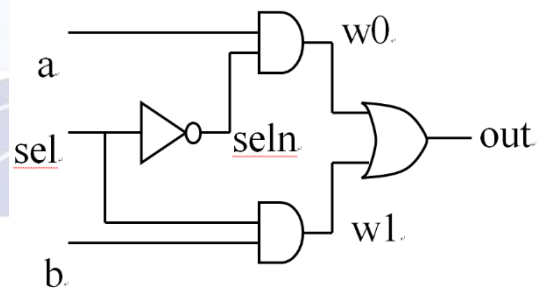
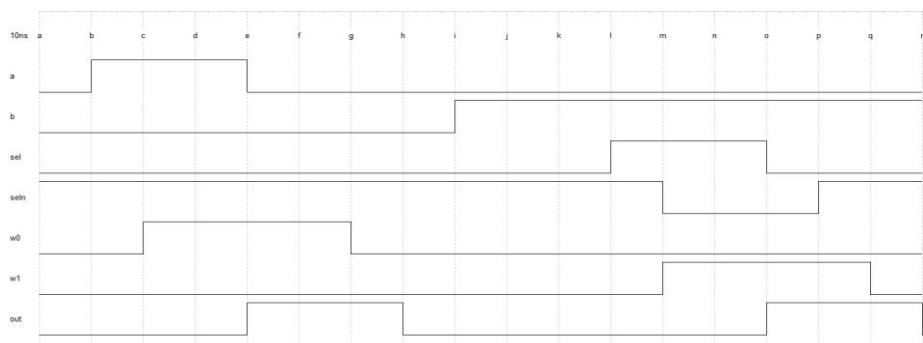


Figure 1.

Ans:



7. Please describe the difference between three macro.

1pt	2pt	1pt
soft macro	Synthesizable RTL	1. Portable and Editable. 2. Unpredictable in terms of performance, timing, area, or power. 3. IP protection risks.
firm macro	Netlist format	1. Performance optimization under a specific fabrication technology 2. Need not synthesizing (sometimes it's time wasting)
hard macro	Hardware (LEF, GDS2 file format)	1. Specifies the physical pathways and wiring (proved under specific tech.) 2. Moving, rotating, flipping freedom but can't touch the interior (APR)